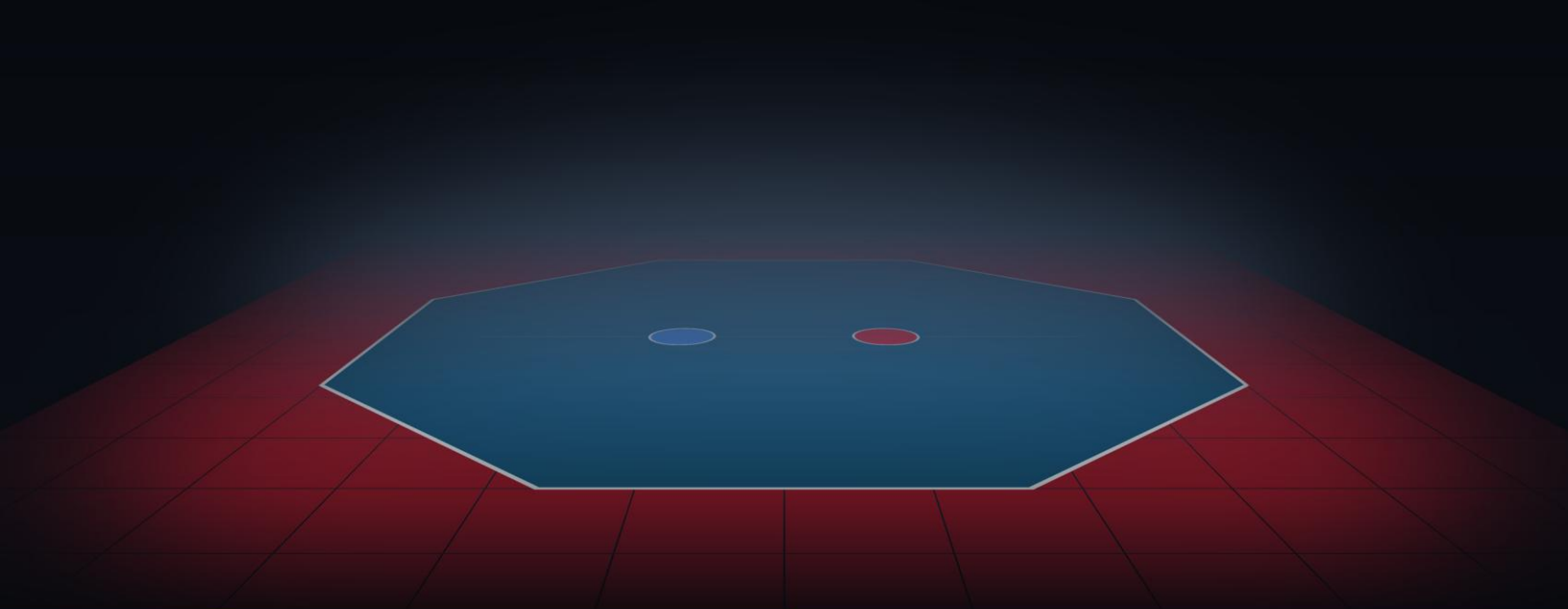




BCIT CAPSTONE · 2026–2027

Open Stance

겨루기 Gyeorugi · Game Design Document

Real taekwondo sparring, one on one: read the distance, choose your kick with a flick of the Skill Stick, and land it before your opponent does.

VERSION	0.4 draft · updated 2026-09-19
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INSTITUTION	British Columbia Institute of Technology
ENGINE	Unreal Engine 5.8 · DirectX 12
PLATFORM	PlayStation 5 and Xbox Series X S · Windows PC (development)
STATUS	Pre-production — draft, not yet reviewed

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1 Overview

Open Stance captures the rhythm and decision-making of real taekwondo: distance, timing, stance, guard, counters and scoring. It starts as a sparring game built around motion capture of real practitioners, and grows into the home of competitive taekwondo sparring: tournaments, team events, online competition and para taekwondo. Open Stance is only about sparring; forms (poomsae) and breaking are out of scope.

This main page states what the game is, why it is worth playing and building, and where the project stands. Detailed design starts at section 2; the Contents lists every section.

Phases. Everything in this document is part of 1.0, the game this capstone delivers: the First Half builds the sparring core, the Second Half adds Career (1.14). A **[Future]** tag marks the exceptions: expansion planned for after 1.0.

1.1 Working Title

Open Stance: Gyeorugi. In taekwondo sparring, an open stance is the position where the two fighters lead with opposite legs, so their trunks face each other and the roundhouse kick is open. Gyeorugi (겨루기) is the sport's own word for sparring. The title tells the player that the game is about stance, position and competitive sparring, presented in taekwondo's own terms.

1.2 Current Status

Status on 2026-09-19: pre-production, in the proposal phase of the First Half. The design is a draft that has not been reviewed, and no gameplay prototype exists. The character and venue art pipeline works. The project is on schedule for the October 2026 proposal.

Report	Items
Complete	Preliminary project plan complete. Controller layout fixed on 2026-09-14 (section 7). Base fighter: MetaHuman body (175 cm, medium build) in seven wardrobe pieces: dobok jacket and trousers, belt, chest protector, helmet, gloves and foot guards, with Chung and Hong materials; passed a full-body range-of-motion test. Referee roster: six MetaHuman referees in uniform, with Chaos cloth on ties and trousers. Staging levels L_CompetitionArena (8 m octagon mat in a 12 m square) and L_TrainingDojang. Motion capture shots planned.
In progress	BCIT proposal: requirements, research, architecture and feasibility (due October 2026). This document.
Blockers	None.

Report	Items
Debt and major issues	The controller layout leaves several details open, such as which right-stick direction gives a roundhouse or hook kick and what Active Defense Mode does (18.1). The runtime module Source/OpenStance has no gameplay code yet, so the Skill Stick stays untested until Milestone 1. Stamina, movement and hit-reaction values are starting values that have not been playtested. Second Half dates are not published.

1.3 Concept Statement

Real taekwondo sparring, one on one: read the distance, choose your kick with a flick of the Skill Stick, and land it before your opponent does.

A match lasts 5 to 8 minutes and one exchange lasts 1 to 3 seconds. One player spars the AI, or two players spar on one console.

1.4 Genre

3D one-on-one sports fighting game. From fighting games, Open Stance takes committed attacks with startup, active and recovery phases, reads, fakes and counters. From sports simulation, it takes a real ruleset: World Taekwondo scoring, electronic protectors, penalties and ranks. There is no health bar and no combo string; matches are won on points, as in the real sport.

1.5 Target Audience

Audience	Who	Games they play	What they want
Fighting-game players	Ages 16–35, controller players	Street Fighter 6, Tekken 8	Depth through spacing and timing rather than long combo strings; a high skill ceiling; local versus
Combat-sports game players	Ages 16–40	EA Sports UFC 5, Undisputed	A realistic sport with simulation controls and a career
Taekwondo practitioners	Athletes, coaches and club members, age 13 and up	Taekwondo Grand Prix, or none	Their sport shown accurately: real rules, techniques and etiquette
Martial arts schools [Future]	Clubs and coaches	—	Tactics teaching and remote sparring leagues

Motivations: mastery (a fair, readable system where skill decides the match), competition (winning on points against a friend or rival) and authenticity (real technique and rules).

ESRB target: E10+ (Everyone 10+), Mild Violence. Contact is full but sporting: no blood, no injuries shown and no taunts. The ESRB assigns the final rating; online interactions are not rated.

1.6 Concept and Unique Selling Points

A match is a series of short exchanges on an 8 m octagonal mat. Each fighter bounces in guard and manages distance with the left stick. The right stick, the Skill Stick, chooses the kick: up for a side or cut kick, left or right for a roundhouse or hook kick, and quarter and half circles for axe, spinning and tornado kicks. Holding RT kicks with the back leg, holding RB aims at the head, and holding LB fakes the attack. Hits score the way the electronic scoring system does: 1 point for a punch, 2 for a body kick, 3 for the head, 4 for a spinning body kick and 6 for a spinning head kick (8.3). Stamina limits how often a fighter throws big techniques. Players learn in the training dojang and spar the AI or a friend; Career then takes a created fighter from white belt at a local club to a world final, unlocking techniques rank by rank. The look is realistic: MetaHuman fighters in white doboks and blue or red gear, animated from motion capture of real practitioners.

Unique selling points

- **The Skill Stick.** Kicks are right-stick gestures, and two held modifiers choose the leg (RT) and the target (RB), so a small set of gestures covers the sport's main kicks at body and head height.
- **Real sparring, real scoring.** World Taekwondo rules: 2-minute rounds, electronic protector scoring, 1 to 6 point techniques, gam-jeom penalties, and a 12-point gap.
- **Distance and commitment decide matches.** Attacks never home in from out of range, and every technique can be punished on recovery (1.7).
- **Motion captured from real practitioners.** Techniques, stance switches, reactions and falls come from taekwondo athletes.
- **Respect for the sport.** Real ranks, competition colours and etiquette, and nothing that affects a match is ever sold (1.7).

Prototype: none yet. The Milestone 1 proof of concept and its video link here when complete (section 19).

1.7 Design Pillars

Pillar	Meaning
Taekwondo first	Techniques and movement that work in real sparring. Nothing is added because it looks flashy in other fighting games.

Pillar	Meaning
Distance matters	Attacks don't magnetically connect from unrealistic range. Small spatial correction is fine; bad distance still misses.
Commitment matters	Startup, active and recovery phases create counter openings. Big techniques are big risks.
Score and stamina are separate	Score decides who is winning. Stamina decides what the fighter can execute well right now.
Animation serves gameplay	Mocap is cut into reusable states, transitions and reactions that gameplay drives.
Respect the sport	Real ranks, real competition colours, real skin, hair and eye colours. Skill decides matches, never purchases.

1.8 Player Experience

Question	Answer
Who is the player	A taekwondo athlete at a competition, in dobok and full sparring gear. In Career the same created fighter grows from a white-belt beginner to a black-belt international competitor.
Setting	The present-day sport: a training dojang and competition venues. There is no scripted plot and no fantasy setting (6.1).
Fantasy	Being a sharp, composed sparring athlete: reading an opponent, landing the head kick at the right moment and winning under scoreboard pressure.
Emotions	Tension while circling at kicking distance; the snap of a clean score; pressure when behind on the clock; pride in a disciplined win; respect at the bow.
Length	A session: 20 to 40 minutes of lessons and matches. A Career season: 3 to 5 hours, with an estimated 30 hours to reach 3rd dan (level 40, 5.2). Match and round lengths: 2.4.
What keeps them playing	Every exchange is a new read, the score and clock change the right decision, and improvement shows as more points from fewer, better techniques.
Major phases	1. Learn: Training lessons for movement, the Skill Stick and scoring (first 15 minutes). 2. Compete: Sparring against the AI and local friends. 3. Master: harder AI, spinning and jumping kicks, clean matches without penalties. Career and Online [Future] extend Compete and Master (section 2).

1.9 Key Moments

Moment	When	What happens
First score	Minutes 1–5	In a Training lesson the player pushes the right stick  sideways. The lead leg throws a roundhouse kick into the protector, the scoreboard shows +2 and the protector sound confirms it.

Moment	When	What happens
First read	First AI match	The AI advances; the player dashes back B and meets its approach with a side kick  . Timing and distance, not button speed, won the point.
The wall	A harder AI or a first match against a friend	The opponent cut-kicks every approach and the player falls 4 points behind. Fake attacks LB and switching sides A get the player past the cut kick.
The comeback	Final round, last 10 seconds	Down by 3, the player fakes an attack, then throws a spin hook kick RT +  for 6 points as the clock runs out.
The podium	Career: first national championship	In a black dan dobok, the fighter bows to the rival who beat them two seasons earlier, then takes gold.

1.10 Art, Sound and Music

Visual style. Realistic and clean, like a broadcast of a World Taekwondo event. Fighters are MetaHuman bodies in white doboks with blue (Chung) or red (Hong) protectors and helmets. Arenas use bright, even light so limbs and scoring zones stay readable. The Chung-blue and Hong-red convention runs through the mat, gear, scoreboard and HUD, so the player always knows which side is which. The training dojang is warmer: daylight, wood, and walls of kicking pads.

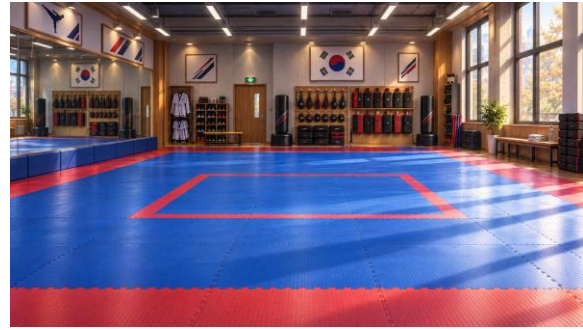
Reference images (Resources/References) set tone and layout. They are not used in the game.



Fighter: dobok, Chung protector and helmet, gloves and foot guards (front, side, back)



Competition arena: octagon mat, scoreboards, officials' tables, coach chairs



Training dojang: kicking pads, heavy bags, mirrors



Referee uniform

Sound. Sound carries information: the protector impact changes with the points scored (1 to 6), feet squeak and slap on the mat, and the referee calls commands in Korean with English subtitles. Crowd size and reverb grow with the venue, from a quiet club night to a full arena.

Music. Music plays in menus, loading screens and round breaks only; rounds have no music, so impacts and footwork stay readable. Style: modern electronic tracks with Korean percussion. Whether music is commissioned or licensed is unresolved (18.1).

Asset lists. A complete art and audio asset list does not exist yet (18.1). Presentation details: section 13.

1.11 Current Target Platform

Item	Target
Platform	PlayStation 5 and Xbox Series X S
Development platform	Windows 10 and 11, 64-bit PC. 1.0 is developed and demonstrated here; a console release needs a licensed developer account and devkits that the project does not have (18.1).
Input	Controller only: one per player, two for local versus. There is no keyboard or mouse play.

Item	Target
Engine and graphics	Unreal Engine 5.8; DirectX 12 on Xbox and on the Windows development build, the PlayStation 5 renderer on PS5
Performance	60 fps during matches: 1920 × 1080 on the Windows development build, 3840 × 2160 on PlayStation 5 and Xbox Series X, 1920 × 1080 on Xbox Series S
Performance verification	Resolution and frame rate measured on the Milestone 3 build (18.1)

A PC release on Steam and cross-play are in section 17.

1.12 Competition

Game	Release	Sales and budget	Description
Taekwondo Grand Prix Hello There Games	PC, 9 October 2018	Not published	The closest direct competitor: taekwondo sparring with World Taekwondo rules and scoring, motion-captured movement, local and online play and customization.
Virtual Taekwondo Refract Technologies with World Taekwondo	VR: PICO 4 and PICO 4 Ultra	Not published	World Taekwondo-endorsed VR sparring with body-worn trackers; the player's body is the controller. Used for the World Taekwondo Virtual Championships (Singapore, November 2024).
Tekken 8 Bandai Namco	PC, PS5, Xbox Series X S; 26 January 2024	About 3 million copies by February 2025	3D fighting game with a taekwondo-based character (Hwoarang); health bars, juggles and combos.
Street Fighter 6 Capcom	PC, PS4, PS5, Xbox Series X S; 2 June 2023	More than 7 million units by June 2026	2D fighting game with a taekwondo-based character (Juri) and simplified Modern controls that widen its audience.
EA Sports UFC 5 EA Vancouver	PS5, Xbox Series X S; 27 October 2023	Not published	Mixed martial arts simulation with stand-up striking, stamina, a career mode and online play.
Undisputed Steel City Interactive	PC early access January 2023; PC, PS5, Xbox Series X S 11 October 2024	1 million copies; about US\$18.8 million raised for development	Licensed boxing simulation built on footwork, stamina and realistic punch selection.
Fight Night Champion EA Canada	PS3, Xbox 360; March 2011	Not published	Boxing game whose analog-stick punching is the closest precedent for choosing strikes with a stick gesture.

What sets Open Stance apart. Taekwondo Grand Prix, the only direct competitor, is PC only, and Virtual Taekwondo needs a VR headset and body trackers. No taekwondo sparring game ships on PlayStation or Xbox; Open Stance targets both. Open Stance differs from both through the Skill Stick, which gives controller players a readable, expressive way to choose kicks. It differs from Tekken 8 and Street Fighter 6 by using the sport's real rules and scoring instead of health bars, and from UFC 5 and Undisputed by bringing their simulation approach to a kicking sport neither covers.

Sources: Steam store and Metacritic (Taekwondo Grand Prix); virtualtkd.gg; Bandai Namco via Game Developer (Tekken 8); Capcom investor relations and Gematsu (Street Fighter 6); EA (UFC 5); Forbes and PlayStation Universe (Undisputed). Checked 2026-09-14.

1.13 Monetization

Phase	Model
1.0	None. 1.0 is a free portfolio and demonstration build.
Future	A commercial release would be premium: a one-time purchase on the PlayStation Store and the Microsoft Store, with optional cosmetic content only (gear styles, dojang items, ceremonies). Never ranks, techniques, attributes or anything that changes a match. The price is unresolved (18.1).

A premium price suits a competitive sport game built on fairness (1.7): every player has the same techniques, and Ranked would give everyone the same attributes. Free-to-play progression would put purchases between the player and techniques, which the balance rules forbid (5.6).

1.14 Scope and Team

Item	Plan
Team	One developer (Joseph Driedger): design, programming, technical art and production. Volunteer taekwondo practitioners perform the motion capture.
Length	About 8 months, in two halves: the First Half (September to December 2026) builds the sparring core and ends in playable demo gameplay, and the Second Half (January to spring 2027) adds Career and finishes the build.
First playable	Milestone 2, December 2026: two local players complete a basic sparring match. This is the demo gameplay that closes the First Half.
1.0 content	1 to 2 fighters, the competition arena and training dojang, 8 to 12 captured techniques, Sparring, Training and local Tournament modes, an AI opponent, and a Career season with training, grading, tournaments, rank-gated technique unlocks and its own venues. Online play, online tournaments, challenges, replays and video review are Future work (section 17).
Time to complete	Sparring and Training have no end state. Career runs about 30 hours to 3rd dan, the level 40 cap (5.2), and the world series final (4.2).
Major risks	Capture of fast kicks, animation workload, responsiveness of mocap-driven attacks, Skill Stick complexity and scope growth. Mitigations: 16.4.

2 Player Objectives and Progression

2.1 The Player

The player controls one taekwondo fighter in a one-on-one match, seen from a side-on camera. In Sparring the player picks a fighter; in Career the player creates one.

What the player knows at the start. Nothing about taekwondo is assumed. The first launch opens Training lesson 1, which explains stance, guard, distance and the scoreboard in under 3 minutes and can be skipped. There is no scripted plot; the season calendar, the rivals and the coach's notes carry the story (4.3).

2.2 Goals

Horizon	Goal	Shown by
Exchange: 1–3 s	Score a technique, or make the opponent miss and counter	Score pop-up and sound
Round: 2:00	Lead on points at time, or reach a 12-point gap	Scoreboard and clock
Match: 5–8 min	Win 2 rounds	Round indicators and results screen
Session	Finish lessons, raise drill grades, beat a harder AI	Lesson list, grades, difficulty list
Season	Pass grading exams and qualify for higher event tiers	Career calendar and ranking points
Career	Win the world series final; enter the hall of fame [Future]	Career record

Progression is linear only where rank gates it: Career event tiers and technique unlocks follow rank (5.3, 5.5). Everything else is the player's choice: which events to enter, what to train and how to fight.

2.3 Moment-by-Moment Gameplay

READ DISTANCE → SET UP → COMMIT → SCORE OR MISS → RESET

- **Read distance (0.5–2 s).** Both fighters bounce in guard just outside kicking range. The player steps and angles with the left stick  and switches sides with , watching the opponent's lead leg and weight.
- **Set up.** A fake attack (hold ) , a guard change on the D-pad  or a dash  draws a reaction.

- **Commit.** A Skill Stick gesture throws the technique, with **RT** for the back leg and **RB** for the head. Its startup, active and recovery phases (MoveData, 16.3) decide whether it lands before the opponent's counter.
- **Score or miss.** A valid hit on the protector or helmet scores on the HUD with a sound matched to the points. A miss leaves the attacker in recovery, open to a counter.
- **Reset.** Fighters return to guard; the referee calls kal-yeo to separate a clinch. The next read begins.

Controller layout: 7.1. Techniques and inputs: section 9.

2.4 Core and Outer Loops

Loop	Length	Cycle	Why it engages
Exchange (core)	1–3 s	See 2.3	Constant small decisions with immediate, readable feedback
Round	2:00	Exchanges → score changes → adjust risk to the score and clock	The clock and point gap change which risk is right
Match	5–8 min	Round → 1:00 break → round	Adapting to an opponent's habits between rounds
Session	20–40 min	Train a technique → spar → review results → train	Visible improvement in grades and results
Career week	10–20 min	See 4.1	Rank, rivals and events give long-term goals

Expected emergent play: players develop styles (counter fighter, pressure fighter, head-kick hunter) from one shared move list; they condition an opponent with repeated fake attacks before throwing the real kick; and they take late-round risks when behind. The loops contain no monetization points (1.13).

2.5 Interactivity

Type	In Open Stance
Action and feedback	Stick gestures and held modifiers map directly to techniques, legs and targets. The score appears the moment a hit registers, with a protector sound and reaction matched to the points.
Short-term cognitive	Reading distance, stance (open or closed) and the opponent's guard; choosing body or head, fake or commit, in under a second.
Long-term cognitive	Learning an opponent's habits across rounds; managing stamina and the point gap over a match; planning training and events in Career.
Emotional	Tension at kicking range, relief at a block, pride in a clean spinning kick, respect at the bow.
Social	Local versus on one console; online casual and ranked [Future] ; clubs and team events [Future] .

Type	In Open Stance
Cultural	The sport's etiquette, Korean commands and ranks; a way for practitioners to share their sport with people who have never trained.

2.6 Progression Through the Game

TRAINING LESSONS → SPARRING VS AI → LOCAL VERSUS → CAREER

The player starts with the six Training lessons (section 3), then spars the AI at rising difficulty and plays friends. Outside Career every technique is available from the start. The Second Half adds Career, with rank-gated progression and technique unlocks (sections 4 and 5); Online play is Future work (section 12). Key moments along this path: 1.9.

3 Game Modes

3.1 Play

Sparring. A single match with no progression attached.

- Player vs player (local) or player vs AI.
- Settings: ruleset options (8.2), arena, AI difficulty, fighters.
- All techniques unlocked; Equal Terms attributes by default (5.6).

Tournament. Single-elimination brackets of 8, 16 or 32, played locally. AI fighters fill every slot the players do not take, so one player can run a full bracket alone. Several local players can enter one bracket.

Team Championship [Future]. Club or national teams in a tag-team format modelled on WT team events: fighters rotate in and out during a bout, and a shared team score decides the match. Up to three local players per team.

Challenges [Future]. Short scenarios with goals, such as “Win the round from 5 points down with 30 seconds left” or “Score a spin hook kick right after switching sides”. Medals unlock cosmetics.

3.2 Progress

Training (도장 dojang)

- **Lessons:** six guided tutorials: movement; Skill Stick kicks; back leg, head and fake modifiers; switching sides; guards and active defense; stamina.
- **Drills:** kicking paddles, reaction targets and switch-side timing, with grades.
- **Free practice:** dummy or AI partner, record and playback, input display, debug overlay.

Career. Take a created fighter from white belt at a local club to world-level events. See section 4.

3.3 Other Sparring Formats

Para taekwondo [Future]. Para gyeonggi under WT Para Taekwondo rules (for example, no kicks to the head), with animation captured with para athletes and classification-accurate fighters. Developed with para practitioners, not approximated.

3.4 Connected

Online [Future]

- **Casual:** quick match and private rooms.

- **Ranked:** seasons with a skill rating; Equal Terms always on. See section 12.

Replays [Future]. Saved matches replayed from inputs and scoring events, with free camera, slow motion and clips of scoring exchanges. Photo mode and sharing **[Future]**.

Broadcast [Future]. Spectator and commentator tools for community tournaments: director camera, live scoreboard overlays, instant replay and bracket display.

4 Career

Career follows one created fighter through seasons of training, grading and competition. A season calendar sets training slots, grading dates and the events the fighter enters; tournaments run in several formats (4.2), and ranking points decide which tier opens next. Techniques unlock by rank and are learned in a drill (5.5), so the move list grows with the fighter. There is no scripted plot; rivals, gradings passed and titles won are the record (6.1). Career menus show the fighter out of match gear: in dobok and belt, at the club, on the podium.

4.1 Weekly Loop

PLAN THE WEEK → TRAIN → GRADE OR COMPETE → REVIEW → LEVEL UP

- **Plan:** a season calendar of training slots, rest days, grading dates and the events the fighter has entered.
- **Train:** drills and sparring earn XP and attributes, and a newly unlocked technique is learned in a drill before it can be used (5.5). Training without rest lowers readiness for the next event.
- **Prepare:** before an entered event the coach scouts the bracket and opens drills aimed at one opponent's habits; the last week is a taper that trades XP for readiness.
- **Grade:** belt grading exams gate rank (5.3).
- **Compete:** events award ranking points, XP and unlocks.
- **Review:** points by technique, stamina use and penalties, with a coach's note pointing to a drill.

4.2 Event Tiers

Event names are fictional, so no real competition is licensed. Tiers differ in format as well as size. A club night is a round robin. The opens and championships are single-elimination brackets, seeded on ranking points from tier 4 up. The world series final splits its eight qualifiers into two pools before the knockout.

Tier	Event	Entry	Format	Bracket
1	Club sparring night	Any rank	Round robin	4
2	City open	Yellow belt	Single elimination	8
3	Provincial championship	Green belt	Single elimination	16
4	National championship	Blue belt, provincial podium	Single elimination, seeded	16
5	Continental open	Black belt, national ranking	Single elimination, seeded	32

Tier	Event	Entry	Format	Bracket
6	World series final	Top 8 in ranking points	Two pools, then knockout	8
7 [Future]	Quadrennial world games	Qualification over a 4-season cycle	Single elimination, seeded	16

4.3 People

- **Coach:** weekly suggestions, opponent scouting before events, post-match notes.
- **Rivals:** recurring AI fighters with their own styles who rise through the tiers alongside you.
- **Club mates:** sparring partners who unlock drills and friendly matches.

4.4 Long-Term Career [Future]

- **Age divisions:** start as a cadet (12–14, poom rank, helmets with face shields) or junior (15–17) and move up to senior.
- **Seasons and ageing:** multi-season careers with peak years, gradual attribute decline and retirement.
- **Fatigue and minor injuries:** knocks that affect readiness, handled with rest and physio choices; never permanent or graphic.
- **Hall of fame:** retired fighters' records, medals and signature techniques.

5 Progression and Levelling

Two tracks run side by side. Fighter level comes from XP and grows attributes. Rank (belt and dan) is earned at grading exams and gates events, techniques and dobok style. Neither track lets stats beat skill.

- **Match results screen with points by technique**
- **Fighter level, XP, attributes, ranks and technique unlocks (Career)**
- **Ranked seasons and skill rating (Online) [Future]**
- **Multi-season careers and hall of fame [Future]**

5.1 XP Sources

Source	XP	Notes
Match win	300	×1.0–2.0 by event tier
Match loss	150	Losing still teaches
Round won	50	Per round
Technique variety	≤100	10 per distinct scoring technique in a match
Clean match (no gam-jeom)	50	Rewards discipline
Training drill	40–120	By grade achieved
Grading exam passed	500	Once per rank

5.2 Level Curve

XP to go from level L to $L + 1$ is $250 + 75 \times L$. The level cap is 40, and each level grants 2 attribute points.

Level	XP to next	Total XP to reach	Roughly
2	400	325	First session
5	625	1,750	First city open
10	1,000	5,625	Green belt range
20	1,750	19,000	Red belt range
30	2,500	39,875	1st dan range
40	—	68,250	Cap

5.3 Ranks

Belt colour order varies between schools; this is the game's standard. A grading exam is a set of technique trials plus a graded sparring bout, available once the level gate is met.

Rank	Belt	Collar	Level gate	Unlocks
10th–9th  gup	White	White	1	Tier 1 events
8th–7th gup	Yellow	White	4	Tier 2 events
6th–5th gup	Green	White	8	Tier 3 events
4th–3rd gup	Blue	White	14	Tier 4 events
2nd–1st gup	Red	White	20	Advanced drills
1st  dan	Black	Black	28	Tier 5 events, dan dobok
2nd dan	Black	Black	34	Signature technique slot
3rd dan	Black	Black	40	Tier 6 seeding bonus; second signature technique slot

Under-15 fighters hold  poom rank with a red-black belt and collar (cadet careers, 4.4).

5.4 Attributes

Attributes run from 40 to 99 with 70 as baseline. No attribute changes its effect by more than $\pm 10\%$ from baseline, so good timing from a low-stat fighter beats bad timing from a high-stat one.

Attribute	Affects
Power	Impact margin over the trunk registration threshold; reaction severity
Speed	Startup and recovery time of techniques
Flexibility	Head-kick height tolerance
Stamina	Size of the stamina pool
Recovery	Stamina regeneration, including between rounds
Footwork	Step distance and dash cost
Technique	Target zone tolerance and fake-attack cancel window
Composure	Stamina drain under pressure; passivity timer leniency

5.5 Technique Unlocks

In Career, techniques unlock by rank and are then learned through a drill. The back-leg and head modifiers arrive with the first technique that needs them. Every other mode has the full move list from the start (section 9).

Rank	Techniques
White	Movement, guards, punch, side and cut kick, roundhouse kick to the body
Yellow	Active defense, fake attack, dash, switching sides
Green	Roundhouse kick to the head, spin side kick

Rank	Techniques
Blue	Hook kick, axe kick
Red	Spin hook kick, jump
Black	Tornado kick, signature variations

5.6 Balance Rules

- **Equal Terms: all attributes set to 70. The default in Sparring and Tournament; always on in Ranked [Future].**
- Customization never changes attributes or point values.
- Level, rank and techniques are earned in play. They are never sold, in any phase.

6 Game World

6.1 World Fiction

Open Stance is set in the present-day world of competitive taekwondo sparring. There is no invented backstory: the fiction is the sport itself, with its rules, ranks, etiquette and events. Clubs, event names and rival fighters are fictional so no real organization is licensed (4.2). The player meets this world through the venues and the referee and, in Career, through the coach, rivals and calendar. No separate world-fiction document is needed. Career's writing is calendars, rankings, rival records and coach's notes, not a script.

6.2 Venues

Venue	Unreal level	Used for
Training dojang	L_TrainingDojang	Training lessons, drills and free practice. The first area a new player sees.
Competition arena	L_CompetitionArena	Sparring and Tournament matches, and any Career event whose own venue is not built yet.
Club dojang	Planned	Career tier 1, the club sparring night (4.2).
School gym	Planned	Career tier 2, the city open.
Provincial hall	Planned	Career tier 3, the provincial championship.
National arena	Planned	Career tiers 4 and 5, the national championship and the continental open.
World-final stage	Planned	Career tier 6, the world series final.

Players move between venues through menus; there is no traversal between locations. Career sets the venue from the event tier. Sparring and Tournament setup can choose any built venue (7.5).

6.3 Contest Area

Element	Specification
Mat	Blue octagon 8 m across, inside a red safety area; 12 m × 12 m in total (built by Tools/Unreal/build_arenas.py)
Boundary	The octagon's edge. A fighter with both feet over it receives a gam-jeom (8.4).
Start marks	Chung and Hong marks on the centre line, facing each other. Spacing follows the current WT rules (18.1).
Officials	Referee on the mat; judges' and recorder tables and four scoreboards around it, two on each side; a coach chair in the blue and red corners (1.10)
Camera	Side-on combat camera (section 13)

6.4 Movement and Physics

Movement comes from captured footwork, so step lengths and speeds follow real athletes. The values below are starting values for the Milestone 2 prototype; measurements from the Milestone 1 capture replace them.

Movement	Starting value
Bounce in place	2 bounces per second
Step (tap)	40 cm
Advance or retreat (hold)	1.2 m/s
Dash 	1.0 m in 0.3 s
Side step or circle	35 cm per step
Back-leg roundhouse kick reach to the body	0.9–1.2 m between fighters' centres
Jump height 	40–60 cm
Gravity	Unreal default, 980 cm/s ²
Fighter collision	Capsule radius 35 cm; fighters never overlap, and chest-to-chest contact while both fighters press forward starts a clinch

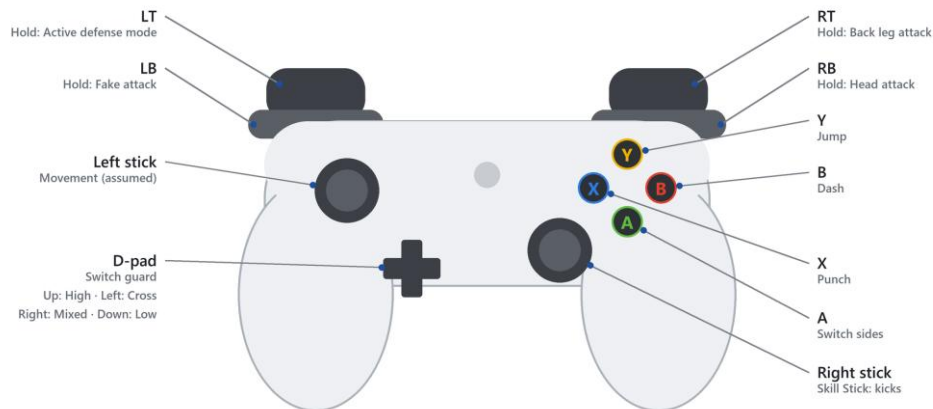
6.5 Easter Eggs

None planned.

7 User Interface

The controller layout is below. In menus, A confirms and B goes back.

7.1 Controller Layout



Inputs are written in Xbox button names; the PlayStation equivalents map one to one. The left stick is not assigned yet, so movement is assumed (18.1).

Input	Function
 Left stick	Movement (assumed)
 Right stick	Skill Stick: kicks (7.2)
 LT Hold	Active Defense Mode
 RT Hold	Back Leg Attack
 LB Hold	Fake Attack
 RB Hold	Head Attack
 Y	Jump
 X	Punch
 A	Switch Sides
 B	Dash
 D-pad	Switch Guard (7.3)

7.2 Skill Stick Moves

Move	Input	Notation
Body Side / Cut Kick		RS Up
Roundhouse and Hook Kick		RS Left/Right
Axe Kick	 then 	RS Left/Right Quarter Circle Forward, Down
Spin Side Kick	 then 	RS Down, Up
Spin Hook Kick	 + 	RT + RS Left/Right Half Circle Down
Tornado Kick	 then hold  +  then release  + 	Y, Hold LB + RS Left/Right Half Circle Down, Release LB + RS Left/Right Half Circle Up

Arc icons show one side; each circle motion works from the left or the right.

7.3 Guards

Guard	Input
High Guard	 D-pad Up
Cross Guard	 D-pad Left
Mixed Guard	 D-pad Right
Low Guard	 D-pad Down

7.4 Input Notes

- **Modifiers combine with the Skill Stick.** Hold  for the back leg and hold  for the head, for example hold  +  for a head roundhouse kick. Kicks without  use the lead leg (to confirm, 18.1).
- **Legs follow stance.** After Switch Sides , the lead and back legs swap and the same inputs still work; the game resolves the physical leg at runtime.
- **Distance still decides.** The input picks the technique; it never pulls the fighter into range (1.7).
- **Accessibility:** hold-to-toggle for the four held modifiers **[Future]** (7.9).

7.5 Screen Flow

TITLE → MAIN MENU → MODE SETUP → FIGHTER SELECT → MATCH → RESULTS

Screen	Contents and flow
Title	Press any button. The first launch goes to Training lesson 1.
Main menu	Play (Sparring, Tournament; Challenges [Future]), Progress (Training; Career), Connected (Online and Replays) [Future], Customize, Options, Quit
Sparring setup	Players (1P vs AI or 1P vs 2P), ruleset (8.2), AI difficulty, arena
Tournament setup	Bracket size (8, 16 or 32), ruleset (8.2), AI difficulty, arena. The bracket view shows results and the next pairing between matches.
Fighter select	Each player picks a fighter. The side (Chung or Hong) is assigned, never chosen.
Match	Intro ceremony, rounds with the HUD (7.6), 1:00 breaks
Pause	Resume, move list, controls, options, restart, quit match
Results	Winner, round scores, points by technique, penalties; rematch, change fighters or main menu
Training	Lessons, drills with grades, free practice (3.2)
Career	Season calendar, training, event entry, grading and post-match review (section 4)
Customize	Fighter creator (section 10) with gear on and off preview
Options	Settings in 7.8

Wireframe mockups for each screen are not drawn yet (18.1).

7.6 HUD

Element	Position	Contents
Scoreboard	Top centre	Chung score in blue on the left, Hong in red on the right, round and clock in the centre, modelled on the PSS display
Penalties	Under each score	Gam-jeom count for that side
Stamina	Under each fighter's name	One bar per fighter; turns amber below 25
Score pop-up	Above the scoring fighter	Points scored (+1 to +6), shown for 1.0 s
Referee subtitle	Lower centre	Korean command and English meaning, for example KAL-YEO: Break
Input display and debug overlay	Left and right edges	Training only: controller icons for the last input, leg, move, target, stamina, hit result
Video review cards [Future]	Beside each score	One card per corner

7.7 Help System

- **Training lessons:** step-by-step tutorials whose prompts show the controller icons and wait for the correct input (3.2).
- **Move list:** in the pause menu and Training; every technique with an input, its icons and points (section 9).
- **Input display:** shows the player's stick gestures and buttons as icons in Training, so a failed input is visible.
- **Loading tips:** one rule or technique tip per loading screen.
- **Coach feedback:** drills recommended from recent match statistics.

7.8 Game Options

Option	Settings
Audio	Master, effects, referee voice, crowd and music volume, each 0–100
Video	Resolution, window mode, quality preset (low, medium, high, epic), frame-rate cap (30, 60, 120, unlimited), V-sync
Controls	Vibration on or off; Skill Stick dead zone; gesture window 0.20–0.40 s (default 0.25 s)
Saving	Settings and created fighters save automatically when changed
Career saving	Autosave after every event and at the end of every week; three career slots
Remapping and hold-to-toggle [Future]	See 7.9

7.9 Accessibility

The game is English only: all text, menus and voice-over. Korean referee commands and rank names are kept as the sport's own terminology, always with English on screen.

- **English subtitles for the Korean referee commands.**
- **Sides shown by position and label, not by colour alone: Chung on the left, Hong on the right, both labelled in text.**
- **Full button remapping, and hold-to-toggle for the four held modifiers (7.4) [Future].**
- **Reduced camera motion and screen-shake options [Future].**
- **Audio cues for incoming attacks and for stamina state [Future].**

8 Core Sparring Systems

The systems that make a sparring match: the systems 1.0 needs. The technique list is section 9 and the data behind each technique is 16.3.

8.1 Systems Overview

System	Intent	Type
Match format (8.2)	Matches play like real WT sparring, so practitioners recognize them and tactics transfer	Player-facing
Scoring (8.3)	Harder, riskier techniques earn more points, as in the sport	Player-facing
Hit detection	Only clean, in-range contact during the active phase scores, so distance matters	Internal
Penalties (8.4)	Stops boundary running, stalling and illegal contact	Player-facing
Referee (8.5, 11.2)	Frames the match with the sport's etiquette and commands	Player-facing
Skill Stick (8.6)	Kick choice is a readable stick gesture; held modifiers choose leg and target	Player-facing
Input recognition and buffer	Turns stick gestures and modifier holds into techniques within a 0.25 s window	Internal
Stance and guards (8.7)	Lead and back legs create open and closed stance matchups; guards change posture	Player-facing
Active defense (8.8)	A held defensive state for answering attacks	Player-facing
Stamina (8.9)	Limits repeated high-commitment techniques without locking controls	Player-facing
Combat state machine	Startup, active and recovery phases create counter windows	Internal
Hit reactions (8.10)	Hits read clearly while the match keeps flowing	Player-facing
AI opponent (11.1)	A sparring partner that uses real tactics	Internal
Video review (8.11) [Future]	Players challenge a missed call, as coaches do	Player-facing

8.2 Match Format

The default ruleset follows World Taekwondo (WT) sparring. Values here are the game's reference; confirm them against the current WT Competition Rules before locking a release, and update them when WT revises its rules.

Rule	Default	Options
Rounds	Best of 3	Single round, best of 3
Round length	2:00	1:00 · 1:30 · 2:00
Break between rounds	1:00	Skippable in local play

Rule	Default	Options
Round win	Most points at time, or point gap	Point gap on/off
Point gap	12 points	Off · 12 · 20
Contest area	Octagonal mat with boundary line	Arena choice
Sides	Chung blue and Hong red	Assigned per match, never chosen as a colour

8.3 Scoring

Hits register the way the electronic protector and scoring system (PSS) does: the trunk protector needs a valid kick above an impact threshold, and the head needs valid contact. In game, a hit registers when the attacking limb's trace reaches the target zone during the active phase with enough impact.

Technique	Target	Points
Punch	Trunk protector	1
Kick	Trunk protector	2
Spinning kick	Trunk protector	4
Kick	Head	3
Spinning kick	Head	6
Opponent penalty (감점 gam-jeom)	—	+1

Tie-breaks follow WT order (higher-value techniques first, then referee decision). Verify the current tie-break order and per-round penalty limit.

8.4 Penalties

A gam-jeom gives the opponent one point, and a fighter who reaches the penalty limit loses the round. Penalties the game detects:

- Stepping both feet over the boundary line.
- Falling down, when not caused by an opponent's penalized action.
- Avoiding or delaying: turning the back, crouching, or not engaging (a passivity timer).
- Lifting the leg to block or impede kicks for more than 3 seconds.
- Kicking below the waist, or striking the head with the hand.
- Grabbing, holding or pushing, detected from contact in a clinch.

8.5 Referee Commands

Command	Meaning	Command	Meaning
CHARYEOT 차렷	Attention, before the bow	KAL-YEO 갈려	Break, fighters separate
KYEONGRYE 경례	Bow	KYE-SOK 계속	Continue
JOON-BI 준비	Ready	KEU-MAN 그만	Stop, end of round
SHI-JAK 시작	Start, the clock runs	GAM-JEOM 감점	Penalty declared

8.6 Skill Stick

The right stick  is the Skill Stick. A direction or motion chooses the kick (7.2); held modifiers change it:  kicks with the back leg,  aims at the head and  turns the input into a fake attack. Every technique with an input is listed in section 9.

Animation flow for every kick (captured, then driven by gameplay):

BOUNCE → CHAMBER → TECHNIQUE → RECHAMBER → LAND → BOUNCE

8.7 Stance and Guards

- **Stances:** Left Lead and Right Lead. Switch Sides  plays a visible switch animation; the fighter is never instantly mirrored. Open and closed stance matchups emerge from positioning.
- **Guards** differ in posture, readiness and transition timing, not stat bonuses. Each guard's posture and strengths are not defined yet (18.1). All four guards ship in 1.0; their inputs are in 7.3.

8.8 Active Defense

Holding  puts the fighter in Active Defense Mode. What the mode does (blocking, parrying, evading, or reading the Skill Stick as defensive directions) is not defined yet (18.1). Candidates are arm blocks to the head and body, a sway, a leg check and stepping out of range.

8.9 Stamina

Stamina is short-term explosiveness. Low stamina lengthens recovery and reduces explosiveness; it never locks out controls.

Action or state	Effect	Starting value
Stamina pool	—	100 (Equal Terms; the Stamina attribute changes it in Career)
Normal footwork and bouncing	No cost	0
Punch 	Small cost	3
Basic kick	Small cost	8
Dash 	Moderate cost	12
Jump or spinning kick	High cost	20
Tornado kick	Highest cost	30
Low stamina	Longer recovery, less explosive techniques	Below 25: recovery phases 30% longer, impact 20% lower
Regeneration	Starts after low activity	15 per second after 1.0 s without spending stamina
Round break	Refill	Back to 100

Starting values for the Milestone 2 prototype, tuned in playtests (18.1).

8.10 Hit Reactions

Severity	Response	Control lost
Minor	Additive torso or head reaction; movement continues	0 s
Normal	Partial or full-body reaction	0.3–0.5 s
Heavy	Captured stagger or loss of balance	0.8–1.2 s
Knockdown	Captured fall, grounded state and get-up	2.5–4.0 s
Physical falls [Future]	Animation-to-ragdoll blend for off-balance falls	As knockdown

8.11 Video Replay Request **[Future]**

Each corner has one video review per match. During a stoppage a player can challenge a missed score or a penalty; the game replays the exchange from scoring data. A successful challenge corrects the score and keeps the review.

8.12 Rulesets **[Future]**

- **Custom rules:** round count and length, point values, point gap, penalty limits, saved as named presets and shareable online.
- **Historical WT rules:** earlier scoring eras for exhibition matches.

- **Other styles:** additional sparring formats (for example ITF-style continuous point sparring) as separate rulesets with their own gear and techniques, if licensing and research allow.

8.13 Epics and User Stories

1.0 work breaks down into these epics. Each story is a testable requirement for its milestone (section 20).

Epic	User stories	Milestone
E1 Skill Stick input	As a player, I push the right stick up to throw a side kick, and hold RT to throw it with my back leg. As a player, I see my last input as controller icons on the input display, so I know why a kick did not come out.	M1
E2 Stance and legs	As a player, after I switch sides with A, holding RT still kicks with my new back leg.	M1
E3 Combat states and hit detection	As a player, a kick that reaches the protector during its active phase scores, and a kick thrown from out of range misses.	M1
E4 Motion capture pipeline	As the developer, I turn one captured roundhouse kick into a retargeted MetaHuman animation in Unreal, by whichever capture method Milestone 1 selects.	M1
E5 Match rules and scoring	As two players, we play a best-of-3 match with 2:00 rounds, WT point values, penalties and a 12-point gap.	M2
E6 Stamina	As a player, repeating spinning kicks drains my stamina and slows my recovery, but I never lose control.	M2
E7 Local versus	As two players with two controllers, we finish a full match on one console.	M2
E8 Production animation	As a player, techniques, transitions and reactions use production mocap and blend without visible pops.	M3
E9 Game flow and UI	As a player, I go from the title screen through setup, a match and results, and back, without restarting the game. As a solo player, I run a local tournament bracket to its final, with AI fighters filling every slot no player takes.	M4
E10 AI opponent	As a solo player, I spar an AI that manages distance, fakes and counters at a difficulty I choose.	M4
E11 Training mode	As a new player, I finish the six Training lessons listed in 3.2.	M4
E12 Career season	As a career player, I plan a training week, enter a tournament, pass a grading exam, and see my rank, ranking points and newly unlocked techniques change.	M4
E13 Balance and polish	As a tester, my beta feedback changes MoveData tuning values without code changes.	M5

9 Techniques

Every technique with an input, its points and the input that performs it; techniques still waiting on an input are in 9.4. Points follow 8.3. Holding RB takes any technique to the head for the head point value, so each table gives the body value first and the head value where the technique can reach it.

9.1 Kicks

Technique	Target · points	Input	Notation
Side kick / cut kick	Body · 2		RS Up
Side kick, back leg	Body · 2	Hold  + 	RT + RS Up
Roundhouse kick	Body · 2		RS Left/Right
Roundhouse kick, back leg	Body · 2	Hold  + 	RT + RS Left/Right
Roundhouse kick, head	Head · 3	Hold  + 	RB + RS Left/Right
Hook kick	Body · 2 / Head · 3	Hold  +  (which direction gives the hook: 18.1)	RS Left/Right; RB for the head
Axe kick	Head · 3	 then 	RS Left/Right Quarter Circle Forward, Down

9.2 Spinning and Jumping Kicks

Technique	Target · points	Input	Notation
Spin side kick	Body · 4 / Head · 6	 then 	RS Down, Up; RB for the head
Spin hook kick	Body · 4 / Head · 6	 + 	RT + RS Left/Right Half Circle Down; RB for the head
Tornado kick	Body · 4 / Head · 6	 then hold  +  then release  + 	Y, Hold LB + RS Half Circle Down, Release LB + RS Half Circle Up; RB for the head

Spinning kicks score the higher trunk and head values (4 and 6) when they register as spinning techniques.

9.3 Punches, Movement and Other Actions

Action	Target · points	Input	Notation
Punch	Trunk · 1		X
Fake attack	—	Hold 	Hold LB
Active defense	—	Hold 	Hold LT
Jump	—		Y
Dash	—		B
Switch sides	—		A
Movement	—	 (assumed)	Left stick (assumed)
Switch guard	—	 High  Cross  Mixed  Low	D-pad: Up high, Left cross, Right mixed, Down low

Punches only score to the trunk protector; punches to the head are penalized.

9.4 Techniques Not Yet Mapped

These techniques have no input assigned yet: push kick, front kick, fast lead-leg roundhouse, double and triple kicks, 360° and jumping back kicks, close-range kicks in the clinch, arm blocks, sway, leg check and clinch actions. They stay out of the move list until they have inputs (18.1); candidates for a larger move library are in section 17.

10 Customization

Fighters are MetaHuman bodies wearing the dobok and sparring gear, refit to each body. Gear is a removable layer: matches show full gear; menus and Career show the fighter in dobok.

10.1 Options

Category	Options	Rule
Body	MetaHuman body types	Sets reach and build
Face	MetaHuman face presets and blends	—
Skin	MetaHuman skin-tone chart	Real skin tones only
Hair, brows, facial hair	MetaHuman grooms	Natural pigment only; no dye
Eyes	Iris colour per eye, pattern	Natural eye colours only; no tints
Side colour	Chung or Hong	Assigned by the match (10.2)
Belt	Rank colour	Earned rank
Dobok collar	White, black; red-black for poom	Follows rank
Dobok back	Name and country code	As printed on competition doboks
Defaults	Preferred lead leg, starting guard	Starting state only
Gear models [Future]	Helmet, protector, glove and foot-guard styles	Visual only; side colours still apply
Dojang [Future]	Club space, banners, club emblem	Emblems moderated online
Ceremony [Future]	Entrance walk, podium celebration, respectful gestures	No taunts

10.2 Side Colours

The chest protector and helmet are only ever Chung blue or Hong red, and the match assigns the side. There is no colour picker for gear, in any phase. Hong red is #C8102E; Chung blue is the protector's baked texture colour.

10.3 Natural Appearance

- **Skin:** MetaHuman skin-tone chart only; no free tint.
- **Hair:** melanin, redness, greying and lightness only; dye colours stay off.
- **Eyes:** natural iris presets (dark brown, brown, amber, hazel, green, blue, grey), set per eye; no tints.

10.4 Unlock Sources

- Rank: belt colours, dan collar, signature technique slot.
- Career milestones: podium poses. Challenge medals and gear styles **[Future]**.
- Any future paid content is cosmetic only, never ranks, techniques or attributes **[Future]**.

11 Game Objects and AI

11.1 AI Opponents

Feature	Description
Difficulty levels	Reaction time, read accuracy, stamina management and technique variety scale by level
Uses real tactics	Manages distance, fakes, guards, counters and the boundary instead of reading inputs
Fighter styles	Counter fighter, pressure fighter, head-kick specialist, cut-kick blocker
Rivals	Named Career opponents whose style evolves between meetings
Adaptive AI [Future]	Learns a player's habits within a match (favourite kick, stance, timing) and adjusts
Shadow fighters [Future]	AI trained from a player's own matches that friends can spar against offline

11.2 Officials

The referee is a non-playable character on the mat. Six referees built from MetaHuman presets (Kelvin, Bo, Jorge, Omari, Walter and Vivian) wear a light-blue shirt, navy tie, beige trousers, brown belt and white sneakers (1.10). Each match picks one at random. The referee speaks only the commands in 8.5.

Situation	Referee behaviour
Match start	Calls charyeot, kyeongrye, joon-bi and shi-jak from between the fighters, then steps back
During play	Moves to keep both fighters in view without blocking the camera
Clinch or stalling	Calls kal-yeo, steps between the fighters, then calls kye-sok
Penalty	Stops play and declares gam-jeom toward the penalized fighter
End of round and match	Calls keu-man; at match end raises the winner's hand

11.3 Equipment and Props

Object	Attributes and values	Behaviour
Trunk protector (PSS)	Side colour Chung or Hong; impact threshold (value unresolved, 18.1)	Registers a kick above the threshold, or a punch, as a score (8.3)
Helmet	Side colour Chung or Hong	Registers valid kick contact to the head
Foot guards and gloves	White	Carry the limb traces that score
Mat and boundary	8 m octagon in a 12 m square	Boundary check on both feet (8.4)
Arena scoreboards	Four screens	Mirror the HUD scoreboard

Object	Attributes and values	Behaviour
Kicking paddles and reaction targets	Height and timing set per drill	Light up to show the target; grade speed and accuracy
Training dummy	Stands still or loops a recorded movement	Free practice target

Technique and object values live in a MoveData data table (16.3) that exports to CSV for review. The table does not exist yet (18.1).

12 Online and Community [Future]

Feature	Description
Rollback netcode	Responsive 1v1 online play; requires deterministic combat simulation (16.3)
Casual and private rooms	Quick match, invite codes, rematch
Ranked seasons	Skill rating; season rank shown on a nameplate, never as a belt
Fair play	Disconnect penalties, report tools, name and emblem moderation
Clubs	Online dojangs with members, internal ladders and club-vs-club team events
Community tournaments	Organizer tools, brackets, check-in, broadcast mode (3.4)
Cross-play	PC and consoles in one player pool

13 Presentation

- **Match intro:** fighters enter in full gear, then the referee's opening commands (11.2).
- **Camera:** side-on combat camera that keeps both fighters and the boundary in frame; 1.5s cuts on spinning-kick scores.
- **Menus:** the player's fighter out of gear in dobok, in a dojang or locker-room setting.
- **Arenas:** the five Career venues listed in 6.2.
- **Commentary [Future]:** play-by-play and colour commentary.

14 Localization

- **Plan:** English only, in every phase. No other language is planned.
- **Korean terminology:** referee commands, rank names and technique names appear in romanization with English meaning, for example KAL-YEO 갈려: Break. They are the sport's terms, not translations.
- **String syntax:** all player-facing text is FText stored in Unreal String Tables, never hard-coded in Blueprints or C++. Values inside strings use FText::Format arguments, for example {Points}.
- **String tables:** ST_UI, ST_Techniques, ST_Referee and ST_Career, in Content/Localization/StringTables (planned; the folder does not exist yet).
- **Key names:** Area.Group.Item, for example UI.MainMenu.Play, Technique.RoundhouseKick.Name, Referee.KalYeo.Subtitle.
- **External files:** none; String Tables export to CSV for proofreading.

15 Tools

Tool	Purpose	Location
MetaHuman Creator	Fighter and referee bodies, faces, skin, hair and eye presets	Unreal Engine 5.8 plugin
Blender pipeline scripts	Rig, split, mask and refit the dobok and gear onto MetaHuman bodies; bake hidden face maps; build referee outfits and environment meshes	Tools/Blender
Unreal Python scripts	Create fighters and referees, import gear as wardrobe items, set up Chaos cloth, run range-of-motion tests, build the arena and dojang levels	Tools/Unreal
Environment texture generator	Mat and venue textures	Tools/Environment
OpenStanceEditorTools	C++ editor module that exposes cloth creation to Python	Source/OpenStanceEdit orTools
Motion capture solver	Turns recorded performance into skeletal animation. Neither the capture method nor the solver is chosen; Milestone 1 compares them	External
IK Rig and IK Retargeter	Retarget mocap onto the MetaHuman skeleton	Unreal Engine
MoveData table	Technique timing, stamina cost, points, target and range, editable without code	Unreal Engine (planned)
Combat debug overlay	Input icons, leg, move, target, stamina, hit result and score in play	In game (planned)
Perforce Helix Core and P4V	Version control for the Unreal project, with binary asset locking	AWS EC2

16 Technical Design

16.1 Stack

Area	Choice
Engine	Unreal Engine 5.8 (source build), OpenStance.uproject
Characters	MetaHuman Creator; dobok and gear as outfit pieces refit per body (Tools/Blender)
Skeleton	MetaHuman body skeleton; gear follows with Leader Pose; mocap retargets with the IK Retargeter
Motion capture	Method not yet chosen. Markerless video capture (MetaHuman Animator, Rokoko Vision) and suit-based capture are both candidates; the Milestone 1 feasibility test decides (18.1, section 19)
Version control	Perforce Helix Core on AWS for the project; Git for tools, docs and portfolio

16.2 Animation Architecture

Need	Approach
Continuous movement	Animation Blueprint and Blend Spaces
Committed attacks	Animation Montages
Hit windows	Anim Notifies driving limb traces and scoring windows
Guards	Layered Blend Per Bone
Mocap retargeting	IK Rig and IK Retargeter
Foot planting	IK correction
Range alignment	Limited Motion Warping
Minor reactions	Additive or layered reactions

16.3 Data and Simulation

- **MoveData:** animation, input gesture and modifiers, startup / active / recovery timing, stamina cost, point value, target, range, reaction properties. Separate from animation assets so placeholders swap for final mocap without changing combat logic.
- **Fighter runtime state:** stance, guard, lead and back leg, held modifiers (active defense, back leg, fake, head), combat state, stamina, score, match state.
- **Combat states:** Idle/Movement, Attack Startup, Attack Active, Attack Recovery, Active Defense, Hit Reaction, Staggered, Knocked Down, Recovering, Match Finished.
- **Fighter profile:** appearance; level, XP, rank, attributes, unlocked techniques and career save.

- **Deterministic combat step [Future]:** fixed-tick combat simulation separated from rendering and animation, so rollback netcode and input-based replays are possible. Combat code in 1.0 is structured so this refactor stays small.
- **Services [Future]:** accounts, matchmaking, ranked seasons, club and tournament backend.

16.4 Major Technical Risks

Risk	Mitigation
Capture struggles with fast kicks and spins	Early feasibility test on a roundhouse kick and a spin hook kick, comparing capture methods and solvers before production (section 19)
Animation workload grows too large	Shared transitions, layering, mirroring and a tightly scoped move list
Mocap looks authentic but attacks feel unresponsive	Gameplay timing separate from performance timing: montages, notifies, limited blending and warping
Physics and ragdoll take too long	Authored mocap falls in 1.0; animation-to-ragdoll blending is Future work (8.10)
Skill Stick gestures are hard to perform or misread	Prototype gesture recognition first, with an input-icon debug display and practitioner playtests
Scope grows beyond 1.0	Online play, online tournaments, challenges, replays, video review and a large roster are Future work (section 17)
A console release needs certification	The game targets PlayStation 5 and Xbox Series X S but is built on Windows, and developer accounts, devkits and certification are not in place (18.1). Windows stays the demonstration platform until they are, and nothing in the design depends on a console-only feature.
Career is a second mode for one developer	Career reuses the sparring core rather than adding systems: the same match, AI, venues and scoring, with a calendar, grading and brackets on top. Its content is data, so it scales down to fewer events without code changes (sections 4 and 5).

16.5 Project Structure and Naming

Path	Contents
Content/Characters/Fighters	Base fighter, gear wardrobe items, hidden face maps
Content/Characters/Officials	Referees and their outfits
Content/Environment	Materials, meshes and textures
Content/Maps	L_CompetitionArena, L_TrainingDojang
Source/OpenStance	Runtime game module: combat, match, input, AI
Source/OpenStanceEditorTools	Editor-only C++ helpers
Tools/Blender, Tools/Unreal, Tools/Environment	Pipeline scripts (section 15)

Path	Contents
Resources	Source art: models, references, textures
Docs	Design documentation

Asset prefixes follow the Unreal conventions already used in the project: L_ levels, MH_ MetaHumans, SK_ skeletal meshes, M_ materials, MI_ material instances, T_ textures, WI_ wardrobe items. Planned: DT_ data tables, AM_ animation montages, ABP_ animation blueprints, BS_ blend spaces, ST_ string tables.

16.6 Data Formats

- **MoveData:** an Unreal Data Table of a C++ row struct, one row per technique; exports to CSV.
- **Saves:** settings and fighter profiles as Unreal SaveGame objects; Career slots use the same format.
- **Animation and gear imports:** FBX in centimetres (UnitScaleFactor 1), matching the MetaHuman export.
- **Replays [Future]:** input and scoring-event streams replayed by the deterministic combat step (16.3).

16.7 AI and Autonomous Systems

The AI opponent (11.1) sends the same inputs as a player and uses the same MoveData; it never reads the player's inputs. Planned structure: a State Tree picks a tactic (keep distance, pressure, fake, counter) and a utility score picks a technique from distance, stamina, score and clock. Difficulty scales reaction time from 450 ms at the easiest level to 180 ms at the hardest (starting values), read accuracy, stamina management and technique variety. The game needs no procedural world, no pathfinding beyond the mat and no economy.

16.8 Test Cheats

Unreal console commands, in development builds only.

Command	Effect
os.InfiniteStamina 0 1	Stamina never drains
os.SetScore <Chung> <Hong>	Sets the score
os.SetClock <seconds>	Sets the round clock
os.AIMode idle block mirror normal	Freezes or scripts the AI
os.ShowHitboxes 0 1	Draws limb traces and target zones
os.CombatDebug 0 1	Shows the combat debug overlay with input icons
slomo <scale>	Unreal time dilation, for example slomo 0.25

Command	Effect
os.SetRank <rank>	Sets Career rank and its unlocks

17 Ideas and Expansions

Everything beyond 1.0. Throughout this document a **[Future]** tag marks a deferred feature where it belongs in a system, so each system reads as a whole; this section summarizes them.

1.0: What this capstone delivers	Future: Every sparring format
Sparring, Training and local Tournament Career: calendar, training, grading, career events Levelling, ranks, attributes, technique unlocks AI opponent, competition arena and training dojang Career venues for event tiers 1 to 6	Online casual and ranked, rollback netcode Challenges and Replays Video review; animation-to-ragdoll falls Larger move library, including the techniques in 9.4 Accessibility options: remapping, reduced motion, audio cues Team Championship, para gyeorugi, cadet and junior careers Clubs, community tournaments, broadcast Adaptive and shadow AI PC release and cross-play

17.1 Later Platforms

- **PC release on Steam, with keyboard play and full remapping.**
- **Cross-play across PlayStation, Xbox and PC.**

18 Unresolved Questions

Open questions are listed here. When one is resolved it moves to 18.2 with a link to the section that answers it; it is never deleted.

18.1 Open

Controls

- Which of RS Left and RS Right gives the roundhouse kick and the other the hook kick, and whether left and right are relative to the opponent or the screen.
- Whether kicks without RT always use the lead leg.
- The spin hook kick lists RT: whether RT is required for it or only selects the back leg.
- The tornado kick holds LB, which is also Fake Attack: how the game tells the two apart.
- What Active Defense Mode (hold LT) does.
- The posture and strengths of the High, Cross, Mixed and Low guards.
- Inputs for movement (left stick assumed), pause and menus, and whether L3 and R3 are used.
- Inputs for the techniques in 9.4, or whether they are cut.
- Playtest the Skill Stick gestures and the 0.25 s gesture window.

Rules and tuning

- Current WT tie-break order and per-round penalty limit.
- Verify WT values: start-mark spacing, the 3-second leg-lift limit, passivity timer length, and the trunk protector impact threshold.
- Number of AI difficulty levels and their tuning values (16.7); AI complexity for 1.0.
- Tune starting values after Milestone 2 playtests: movement (6.4), stamina (8.9), hit reactions (8.10).
- Final move count for 1.0.
- Whether the reach advantage of a taller body needs balancing now that there are no weight classes, or whether Equal Terms and target-zone tolerance already cover it (5.6, 10.1).

Production

- Motion capture method and solver: markerless video capture (MetaHuman Animator, Rokoko Vision) or a suit-based system. Milestone 1 decides (section 19).
- Amount of Motion Warping.
- Natural eye-colour preset values (iris chart sampling in the editor).
- Wireframe mockups for every screen (7.5).
- Create the art and audio asset spreadsheet (1.10) and the MoveData table (16.3).

- Music: commissioned or licensed (1.10).
- Per-platform performance targets, in resolution and frame rate, measured on the Milestone 3 build (1.11).
- Second Half dates once BCIT publishes the 2027 schedule.
- Console developer accounts, devkits and certification for PlayStation 5 and Xbox Series X|S; 1.0 runs on Windows (1.11).
- Belt colour order for release; Career event names.
- Online service provider and the price of a commercial release **[Future]**.
- Licensing path for any real federation branding or events; research partners for para taekwondo **[Future]**.

18.2 Resolved

Question	Resolution	See
Controller layout	Fixed on 2026-09-14	Section 7
Engine	Unreal Engine 5.8 source build	16.1
Version control	Perforce Helix Core on AWS; Git for tools, docs and portfolio	16.1
Scope of the sport	Sparring only: no poomsae, breaking or club-management mode	Section 1
Gear colours	Protector and helmet are only Chung blue or Hong red, assigned by the match	10.2
Appearance	Natural skin, hair and eye colours only	10.3
Character skeleton	MetaHuman body skeleton; gear follows with Leader Pose	16.1
Contest area size	8 m octagon in a 12 m square	6.3
Ragdoll	Authored mocap falls in 1.0; ragdoll blending is Future	8.10
Languages	English only	Section 14
Target platform	Console-first: PlayStation 5 and Xbox Series X S, developed and demonstrated on Windows	1.11
Phases	Two: 1.0 is what this capstone delivers, Future is after it	Main page
Weight classes and divisions	Cut for playability: one open division, no gender split. Body type changes reach and build only	10.1
Head modifier	Holding RB takes any technique to the head, for that target's point value	Section 9

19 Prototypes

No prototypes exist yet. Each one links here when built.

Prototype	Question it answers	Contents	Milestone
Capture feasibility test	Which capture method records fast kicks and spins accurately?	The same back-leg roundhouse kick and spin hook kick through each candidate method and solver; compare supporting-foot pivot, chamber, hip rotation, trajectory, rechamber and landing	M1
Proof of concept	Are the Skill Stick and lead/back-leg model technically feasible?	Greybox arena, mannequin fighters, idle and movement, side kick, roundhouse kick, spin side kick, Skill Stick and modifier recognition, stance-based leg resolution, attack phases, foot traces for head, body or miss, basic scoring and stamina, one retargeted capture	M1
Combat prototype	Does the system create real sparring decisions: faking, countering, distance, body or head?	5–6 techniques, local two-player combat, stamina, scoring, rounds, active defense, switching sides, guards, combat camera, debug overlay with input icons	M2

20 Development Schedule and Milestones

The capstone runs to five milestones across two halves. The First Half (September to December 2026) covers the proposal and Milestones 1 and 2, and ends with playable demo gameplay; the Second Half (January to spring 2027) covers Milestones 3 to 5, Career and the gold build. Exact Second Half dates are added when BCIT publishes the 2027 schedule (18.1).

Period	Phase	Deliverables
September–October 2026	Proposal	Requirements, research, architecture, feasibility, scope and test plan
Fall 2026	Milestone 1	Skill Stick and motion capture proof of concept (section 19)
Fall 2026	Milestone 2	Playable two-player combat prototype: the demo gameplay that closes the First Half (section 19)
December 2026 break	Buffer	Capture preparation, mocap production and cleanup
Winter 2027	Milestone 3	Full combat and animation system with the production mocap library
Winter 2027	Milestone 4	Complete game flow, AI opponent, local Tournament and the Career season: calendar, training, grading, tournaments and rank-gated technique unlocks
Spring 2027	Milestone 5	Beta testing, balancing, polish and requirement verification
Spring 2027	Final	Gold build, working demo and final report

20.1 Milestone Success Criteria

Milestone	Success criterion
Proposal	Approved proposal
M1	Two fighters execute techniques via the Skill Stick; one capture proves the pipeline
M2	Two local players complete a basic sparring match, playable as a demo
M3	Planned combat playable with natural transitions
M4	Startup to finished match against a local player or the AI; a Career season playable from the first club night to a tier 3 final
M5	Requirements verified; ready for demo

Epics per milestone: 8.13. Beyond 1.0: section 17.

21 Change Log

Date	Version	Author	Changes
September 2026	0.1	Joseph Driedger	Preliminary project plan.
2026-09-12	0.2	Joseph Driedger	Earlier draft, built on an older controller layout.
2026-09-14	0.3	Joseph Driedger	Full rewrite, with controller icons and a layout diagram. Adds the main page, player objectives, game world, user interface, game objects, localization, tools, ideas and expansions, prototypes, schedule and change log; starting values for stamina, movement and hit reactions; unresolved questions with a resolved list.
2026-09-19	0.4	Joseph Driedger	Two phases in place of three: 1.0 is what the capstone delivers and Future is everything after it, so only deferred items carry a tag. Online play, online tournaments, video review, challenges, replays and ragdoll blending become Future work; local Tournament against AI opponents stays in 1.0. Retargets the game to console, with Windows as the development platform. Brings Career into the Second Half, with tournament formats, event preparation and rank-gated technique unlocks. Cuts weight classes and gender divisions. Updates scoring to current WT values: a spinning kick to the head scores 6, and spin hook and tornado kicks reach the body as well as the head. Records that holding RB takes any technique to the head, and that the motion capture method is still undecided. Adds a glossary (section 22), controller notation beside the icons in section 9, and console and Career entries in the risk register. Makes the document self-contained and corrects the scope, rank, scoreboard and cross-reference errors listed in 18.2.

22 Glossary

Taekwondo, competition and engine terms as this document uses them. Korean terms are romanized, with the hangul where the sport writes it.

Term	Meaning
Active Defense	A held defensive state on LT. What it does is not yet defined (8.8).
charyeot (차렷)	Referee command: attention, called before the bow (8.5).
Chung (청)	The blue side: blue protector, helmet and corner. The match assigns it; it is never a player's choice (10.2).
closed stance	Both fighters lead with the same leg.
dan (단)	Black-belt grade, counting up from 1st dan (5.3).
dobok	The taekwondo uniform: jacket, trousers and belt.
dojang (도장)	Training hall. The game's Training mode is set in one (3.2).
Equal Terms	The option that sets every attribute to 70, so skill alone decides the match (5.6).
gam-jeom (감점)	A penalty. It awards one point to the opponent (8.4).
gup (급)	Coloured-belt grade below black belt, counting down from 10th to 1st (5.3).
gyeorugi (겨루기)	Sparring: the competitive, full-contact form of taekwondo, and the only form this game covers (1.1).
hogu	The trunk protector. In competition it is electronic and registers scoring kicks (11.3).
Hong (홍)	The red side: red protector, helmet and corner (10.2).
joon-bi (준비)	Referee command: ready (8.5).
kal-yeo (갈려)	Referee command: break, separating the fighters (8.5).
keu-man (그만)	Referee command: stop, ending the round (8.5).
kye-sok (계속)	Referee command: continue (8.5).
kyeongrye (경례)	Referee command: bow (8.5).
Leader Pose	The Unreal Engine method that makes each gear piece follow the body's skeleton (16.1).
MetaHuman	Unreal Engine's photoreal human character system, used for every fighter and referee (16.1).
MoveData	The data table holding each technique's timing, stamina cost, points, target and range (16.3).
open stance	The two fighters lead with opposite legs, so their trunks face each other and the roundhouse kick is open. The game takes its name from it (1.1).
poom (품)	Youth black-belt grade, held under 15, with a red-and-black belt (5.3).
PSS	Protector and Scoring System: the electronic protectors and scoreboard that register hits (8.3).

Term	Meaning
shi-jak (시작)	Referee command: start, and the clock runs (8.5).
Skill Stick	This game's name for the right stick, which chooses the kick (8.6).
WT	World Taekwondo, the sport's international federation, whose competition rules the game follows (8.2).